

WORKFLOW.md

SynChain System Workflow

Overview

SynChain operates as a multi-agent decision intelligence platform that creates a virtual representation of a supply chain and evaluates possible future outcomes before recommendations are generated.

High-Level Workflow

User ↓ Simulation Creation ↓ Digital Twin Generation ↓ Agent Execution ↓ Decision Synthesis ↓ Results Dashboard

Detailed Workflow

Step 1: User Creates Simulation

User enters:

- Product Information
 - Inventory Levels
 - Warehouse Information
 - Demand Information
 - Supplier Information
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Step 2: Digital Twin Initialization

System creates a virtual representation containing:

- Inventory State
- Warehouse State
- Supplier State
- Demand State
- Transportation State

This becomes the Digital Twin.

Step 3: External Intelligence Collection

Agents gather:

- Industry News
 - Market Signals
 - Supplier Information
 - Risk Events
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Step 4: Demand Analysis

Demand Agent evaluates:

- Current Demand
- Historical Trends
- External Signals

Output:

- Demand Forecast
 - Confidence Score
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Step 5: Inventory Optimization

Inventory Agent evaluates:

- Current Inventory
- Forecasted Demand

Output:

- Recommended Stock Level
 - Safety Stock
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Step 6: Logistics Planning

Logistics Agent evaluates:

- Warehouse Capacity
- Product Availability

Output:

- Warehouse Recommendation
 - Distribution Strategy
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Step 7: Risk Assessment

Risk Agent evaluates:

- Supplier Delays
- Inventory Risks
- Market Risks

Output:

- Risk Score
 - Risk Explanation
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Step 8: Scenario Simulation

Scenario Agent simulates:

- Demand Increase
- Demand Decrease
- Supplier Failure
- Logistics Disruption

Output:

- Future Outcomes
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Step 9: Recommendation Generation

Recommendation Agent combines all outputs.

Output:

- Final Strategy
 - Action Plan
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Step 10: Explainability Layer

Explanation Agent generates:

- Decision Reasoning
 - Supporting Factors
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Step 11: Dashboard Presentation

User receives:

- Demand Forecast
 - Inventory Recommendation
 - Logistics Plan
 - Risk Analysis
 - Final Strategy
 - Scenario Results
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Design Principle

Every recommendation must answer:

1. What should be done?
2. Why should it be done?
3. What happens if it is not done?